3 **(a)** Rate of change of angular displacement [B1]

(b) **(i)** By conservation of energy,

Loss in Ep = Gain in Ek

$$mgh = \frac{1}{2}mv^2 - 0 \quad [C1]$$

$$m(9.81)(4.5+5.5) = \frac{1}{2}mv^2 \quad [M1]$$

$$v = 14 \text{ m s}^{-1} \quad [A0]$$

Do not accept use of kinematics equation.

(ii)

$$\omega = \frac{v}{r} = \frac{14}{4.5}$$

$$= 3.1 \text{ rad s}^{-1} \quad [B1]$$

(iii) (Tension – weight) provides for the centripetal force. [B1]

$$F_{net} = ma$$

$$T - mg = m\frac{v^2}{r}$$

$$T - (65)(9.81) = (65)\left(\frac{14^2}{4.5}\right) \quad [C1]$$

$$T = 3469 = 3470 \text{ N} \quad [A1]$$

(iv) <u>Parabolic</u> path with initial horizontal velocity [B1]

4 **(a)** **(i)** $V_{...} = 0.040 \times 80 = 3.2\text{V}$ [M1]